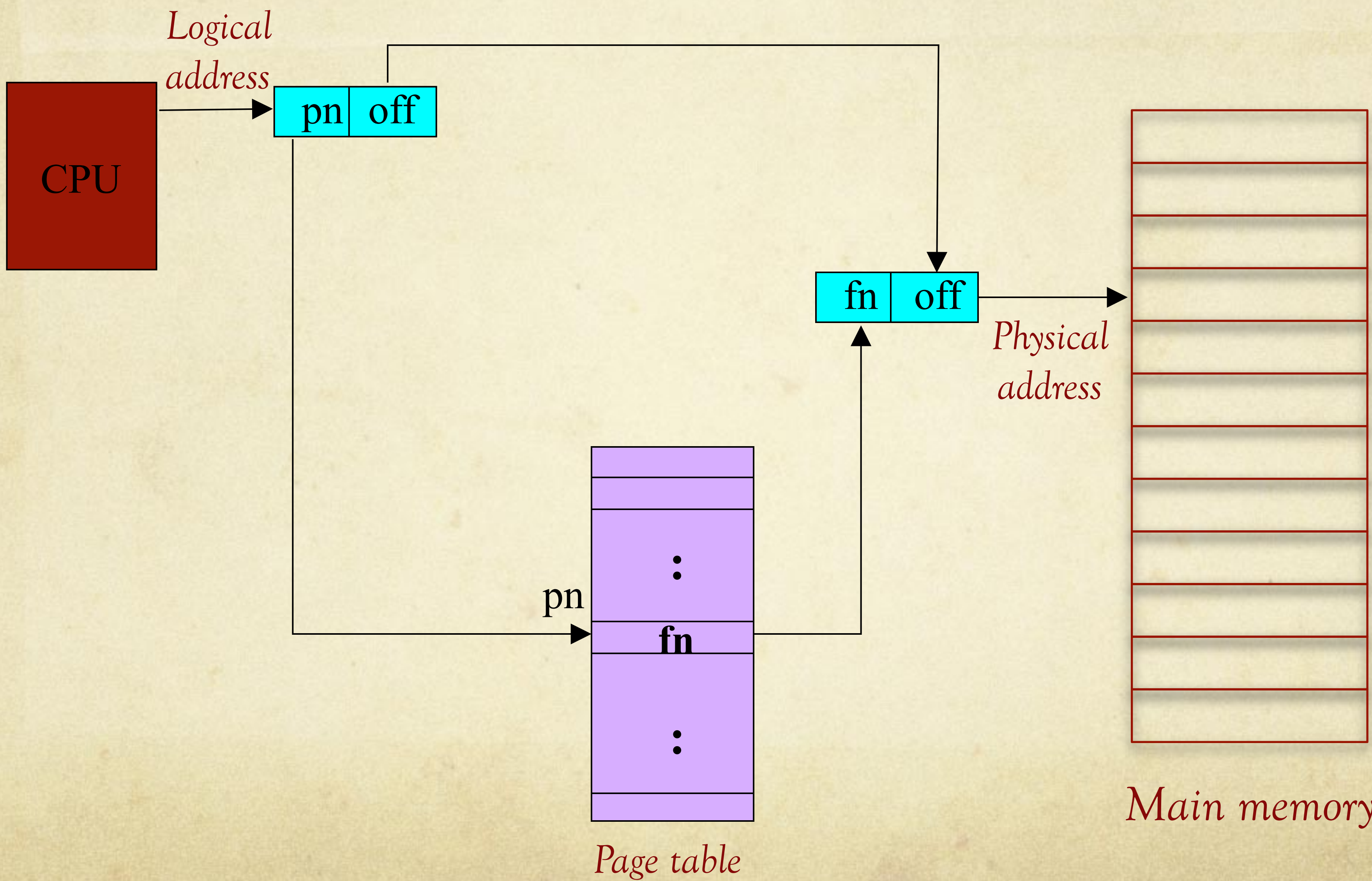
Address translation in paged systems

- *Logical* address includes two parts
  - Page number (pn)
  - Offset within page (off)

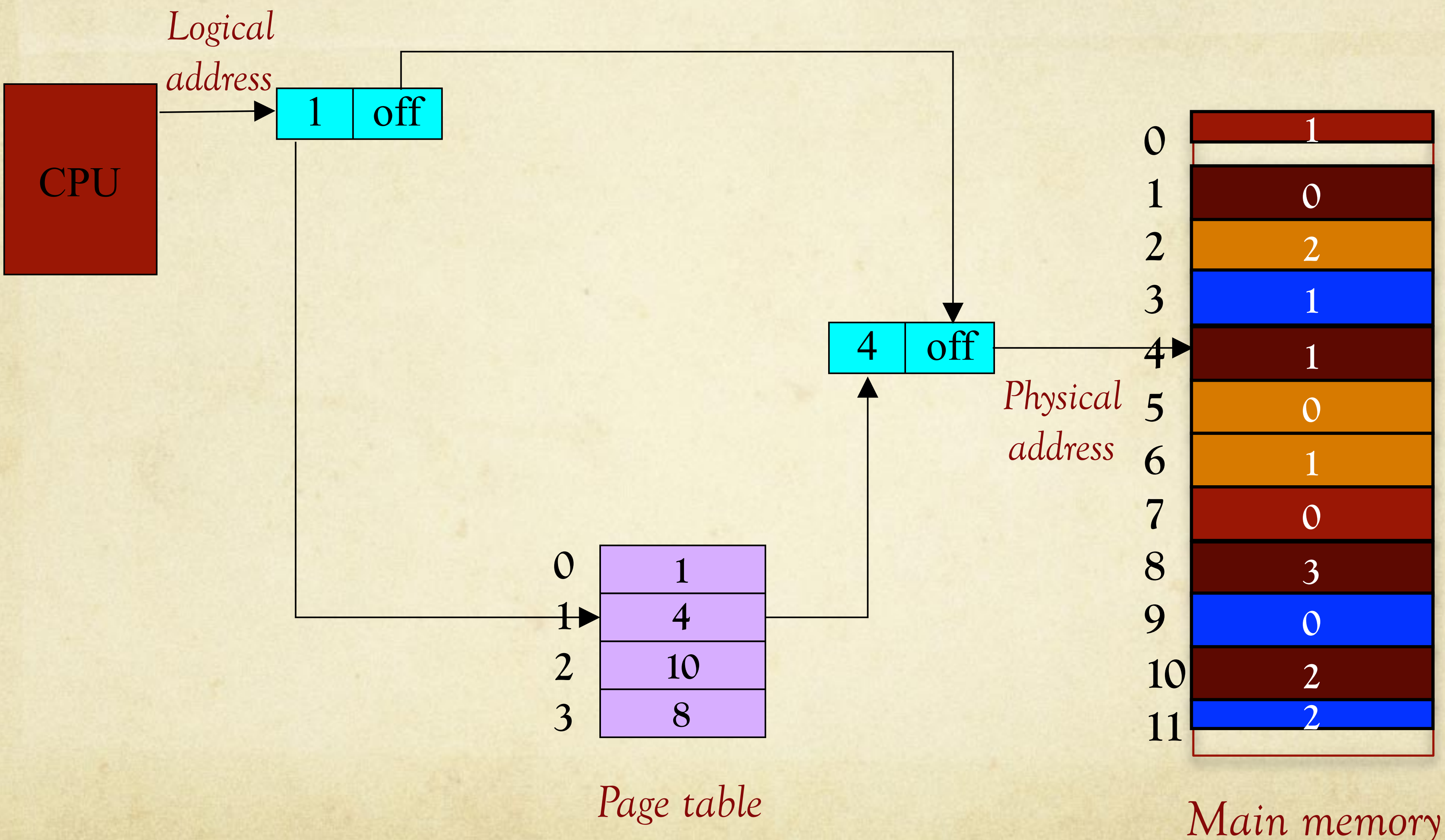
Page table

- *Physical* address includes corresponding parts
  - Page frame number (fn)
  - Offset within frame (off)

Address translation in paged systems



Address translation in paged systems



In a computer ...

- Addresses represented in *binary*
  - Main memory & page frame sizes are powers of 2
  - *Hexadecimal* → more compact
- *Knowledge of number systems & conversions needed*
- Binary → bit values 0 and 1
  - 2 bits: 4 (2^2) combinations (00, 01, 10, 11)
  - 3 bits: 8 (2^3) combinations (000, 001, 010, 011, 100, 101, 110, 111)
  - n bits: 2^n combinations

Memory System

- 32KB (2^{15}) main memory
  - 15 bit physical address
- 16KB (2^{14}) logical address space
  - 14 bit logical address
- 4KB (2^{12}) page frame /page
  - 12 bit offset [*LSB*]
  - $2^{15} / 2^{12} = 2^3$ page frames
    - 3 bit page frame number
  - $2^{14} / 2^{12} = 2^2$ pages
    - 2 bit logical page number

Address translation in paged systems